

PyroGuard AI: Discord Bot

Operational Manual & Slash Command Specification

Near-Real-Time Wildfire Intelligence, Radar Fuel Moisture, & Thermal Hotspot
Monitoring

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Republic of Indonesia Wildfire Intelligence Integration

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Abstract

This document serves as the official reference manual for deploying, configuring, and executing application slash commands via the **PyroGuard AI Discord Bot**. PyroGuard AI enables disaster response authorities (BPBD, BNPB, KLHK, Manggala Agni), forestry rangers, and plantation concession operators to query all-weather Synthetic Aperture Radar (Sentinel-1 SAR), optical canopy indices (Sentinel-2 NDWI), NASA FIRMS thermal anomalies, and machine learning risk forecasts across Indonesia directly within collaborative Discord incident channels.

Contents

1	Introduction and Architecture Overview	2
1.1	Core Architecture	2
2	Prerequisites and Installation	2
2.1	Discord Developer Portal Configuration	2
2.2	Environment Setup	2
3	Slash Command Reference	3
3.1	Command: <code>/risk</code>	3
3.1.1	Execution Mechanics and Algorithms	3
3.1.2	Example Embed Fields	4
3.2	Command: <code>/hotspots</code>	4
3.2.1	Key Metrics Returned	4
3.3	Command: <code>/ask</code>	5
3.3.1	Sample Usage	5
3.3.2	Agent Execution Sequence	5
3.4	Command: <code>/regions</code>	5
4	Ecosystem Specializations in Bot Responses	5
5	Troubleshooting & Common Error Codes	6
6	Command Summary Quick Reference	6

1 Introduction and Architecture Overview

Equatorial wildfire monitoring in Indonesia is frequently impeded by cloud cover, smoke plumes, and remote topography. PyroGuard AI provides a headless, bot-native interface to an autonomous ReAct agent and a quantitative temporal machine learning model.

1.1 Core Architecture

- **Remote Sensing Engine:** Google Earth Engine (GEE) integration querying C-band Sentinel-1 SAR (VV/VH polarization) and Sentinel-2 multispectral imagery.
- **Thermal Hotspot Connector:** NASA FIRMS (VIIRS 375m and MODIS 1km) NRT API coupled with Indonesian Ministry of Environment and Forestry (KLHK) Peatland Hydrological Unit (*Kesatuan Hidrologi Gambut* / KHG) boundaries.
- **Inference Pipeline:** Local ONNX Runtime neural model predicting the Wildfire Vulnerability Index ($WVI \in [0.0, 1.0]$) and 14-day risk trajectories in < 20 ms.
- **Client Integration:** Modern discord.py v2 application tree broadcasting rich interactive embeds, telemetry plots, and tactical recommendations.

2 Prerequisites and Installation

2.1 Discord Developer Portal Configuration

Before starting the bot, ensure the following steps are performed in the [Discord Developer Portal](#):

1. **Create Application:** Register an application named PyroGuard AI.
2. **Acquire Bot Token:**
 - Navigate to **Settings** → **Bot**.
 - Click **Reset Token** and copy the resulting string (~70 characters).
 - Enable **Message Content Intent** under Privileged Gateway Intents.
3. **OAuth2 URL Generator:**
 - Under **Scopes**, select: `bot` and `applications.commands`.
 - Under **Bot Permissions**, grant: `Send Messages`, `Embed Links`, `Attach Files`, `Read Message History`.
 - Open the generated invitation URL to add the bot to your target server.

2.2 Environment Setup

Add the token to your local `.env` file:

```
# .env
DISCORD_BOT_TOKEN=MTIzNDU2Nzg5MDEyMzQ1Njc4OQ.G1xY2z.
  abc123def456ghi789jkl012mno345pqr
GEE_PROJECT_ID=your-gcp-project-id
FIRMS_MAP_KEY=your_firms_map_key
```

Execute the bot launcher:

```
chmod +x run_discord_bot.sh
./run_discord_bot.sh
```

3 Slash Command Reference

PyroGuard AI provides four core application commands accessible by typing / in any permitted channel.

3.1 Command: /risk

The /risk command evaluates the quantitative 14-day wildfire vulnerability of a specified jurisdiction using coupled radar backscatter and weather telemetry.

/risk [region]

Parameters:

- **region** (*string, required, autocompleted*): The name of the target Indonesian regency, province, or national park. Supported options include:
 - Kapuas Regency (Central Kalimantan)
 - Pulang Pisau Regency (Central Kalimantan)
 - Mount Bromo National Park (East Java)
 - Bengkalis Regency (Riau)
 - Siak Regency (Riau)
 - Ogan Komering Ilir Regency (South Sumatra)
 - Baluran National Park (East Java)
 - Kubu Raya Regency (West Kalimantan)

3.1.1 Execution Mechanics and Algorithms

Upon receiving the command:

1. The bot resolves the region geometry and boundary bounding box (W, S, E, N).
2. The 30-day temporal feature vector is calculated:

$$\Delta\sigma_{VV}^0 = \sigma_{VV, \text{wet}}^0 - \sigma_{VV}^0(t) \quad (1)$$

3. In peatland ecosystems, the **Peatland Drying Index (PDI)** is evaluated:

$$I_{\text{PDI}} = 0.50 \cdot f_{\text{SAR}}(\Delta\sigma_{VV}^0) + 0.25 \cdot f_{\text{LST}}(\Delta T_s) + 0.25 \cdot \exp\left(-\frac{R_{14}}{25}\right) \quad (2)$$

4. The ONNX model runs inference in < 20 ms, outputting the **Wildfire Vulnerability Index (WVI)**:

WVI Range	Risk Tier	Color Hex	Discord Embed Color
≥ 0.80	CRITICAL	#DC2626	Red
0.60 – 0.79	HIGH	#EA580C	Orange
0.35 – 0.59	MODERATE	#CA8A04	Yellow / Gold
< 0.35	LOW	#16A34A	Green

3.1.2 Example Embed Fields

The returned Discord embed features:

- **Vulnerability Index:** Numerical score and tier badge.
- **Hydrology / Soil Regime:** Peat water table depth in cm (relative to the -40 cm statutory BRGM limit) or volcanic savanna mineral classification.
- **Remote Sensing Telemetry:** 14-day $\Delta\sigma_{VV}^0$ in dB, foliar NDWI change (%), and thermal anomaly ($^{\circ}\text{C}$).
- **Top Drivers:** Proportional attribution percentage for top 3 risk factors.
- **Tactical Advisory:** Tailored command actions.

3.2 Command: /hotspots

The `/hotspots` command queries active satellite thermal detections with spatial cross-referencing against conservation and peat boundaries.

`/hotspots [region] [lookback_days]`

Parameters:

- **region** (*string, required, autocompleted*): Target jurisdiction.
- **lookback_days** (*integer, optional, default: 3*): Temporal window between 1 and 7 days.

3.2.1 Key Metrics Returned

- **Total Hotspots:** Aggregated VIIRS 375m and MODIS 1km detections.
- **Protected Zone Count:** Detections specifically located within KLHK Peat Hydrological Units (*KHG Fungsi Lindung*) or National Park conservation zones.
- **Fire Radiative Power (FRP):** Cumulative and maximum FRP in Megawatts (MW).
- **Smoke Dispersion Vector:** Indicates direction of downwind atmospheric transport based on 10m wind velocity vectors (u_{10}, v_{10}).

3.3 Command: /ask

The `/ask` command exposes the full autonomous ReAct agent, enabling free-form operational reasoning and situational reporting.

`/ask [query]`

Parameters:

- **query** (*string, required*): Natural language question regarding fire risk, weather, or field actions.

3.3.1 Sample Usage

- `/ask query: "Assess fire risk at Mount Bromo National Park near tourist areas"`
- `/ask query: "What is the peat drying status in Kapuas Regency and are there hotspots?"`

3.3.2 Agent Execution Sequence

When invoked, the agent performs asynchronous reasoning:

1. **Step 0:** Extracts entities and resolves the geographic coordinate boundary.
2. **Step 1:** Invokes `extract_regional_fire_metrics` for 30-day time-series.
3. **Step 2:** Invokes `fetch_active_hotspots` for real-time VIIRS anomalies.
4. **Step 3:** Ingests Open-Meteo forecasts and runs ONNX inference.
5. **Step 4:** Synthesizes a structured tactical intelligence briefing formatted as a Discord message.

—

3.4 Command: /regions

The `/regions` command provides situational awareness on all jurisdictions currently ingested into the monitoring matrix.

`/regions`

Parameters: *None*

Returns an interactive directory containing administrative IDs, province associations, center coordinates, and dominant biome classifications (e.g., Peat Swamp Forest vs. Highland Volcanic Savanna).

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4 Ecosystem Specializations in Bot Responses

PyroGuard AI dynamically adjusts its telemetry analysis and advisories based on the underlying Indonesian ecosystem:

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Attribute	Tropical Peatland (Kalimantan/-Sumatra)	Highland Volcanic (Mount Bromo, East Java)	Savanna
Soil Type	Peat Dome (<i>Kubah Gambut</i>), > 60% Organic	Volcanic Ash / Sand (Regosol), 0% Peat	
Primary Hazard	Subterranean smoldering in dried acrotelm	Rapid surface grass fires (<i>ilalang, pakis</i>)	
Radar Indicator	SAR VV backscatter drop (> 2.5 dB = critical drawdown)	Topsoil moisture desiccation	
Statutory Metric	Water table drop past BRGM -40 cm threshold	Dry spell duration & highland katabatic winds	
Advisory Focus	Canal block re-wetting (<i>sekat kanal</i>), water bombing	Hiking trail bans, firebreaks (<i>sekat bakar</i>), <i>gepyok</i> beaters	

Table 1: Comparative analysis of ecosystem-adaptive bot responses.

5 Troubleshooting & Common Error Codes

Common Operational Errors

- `discord.errors.LoginFailure`: Improper token has been passed (401):
 - *Cause*: An Application ID, Public Key, or OAuth2 Client Secret was placed in `DISCORD_BOT_TOKEN`.
 - *Fix*: Navigate to **Discord Developer Portal** → **Bot** → **Token** → **Reset Token**. Copy the ~70-character bot token into `.env`.
- **Slash commands not appearing in Discord**:
 - *Cause*: Bot was invited without the `applications.commands` scope, or global sync is pending.
 - *Fix*: Re-invite the bot using the OAuth2 URL generator with `applications.commands` checked. Global sync completes automatically upon bot startup.
- **Interaction Timed Out**:
 - *Cause*: Remote sensing computations exceeded 3 seconds before sending an initial response.
 - *Mitigation*: All PyroGuard AI commands implement `await interaction.response.defer(thinking=True)` to permit up to 15 minutes of asynchronous execution.

6 Command Summary Quick Reference

Slash Command	Key Arguments	Primary Output
/risk	region	WVI score, peat water table level, SAR/NDWI deltas, advisory
/hotspots	region, lookback_days	Thermal count, FRP metrics, peatland overlap, smoke vectors
/ask	query	Full ReAct agent multi-step analysis & operational markdown
/regions	<i>None</i>	Directory of monitored provinces, parks, and coordinates

Table 2: PyroGuard AI Discord Slash Command Quick Reference.